PIPELINE AUDIT AND HEALTH REPORT

Transport: airport security checkpoint

Priority passengers use a fast-track lane; everyone else queues for one of two ID desks, a tray belt, then the shortest of three X-ray lanes (which break down now and then). 8% are pulled aside for a secondary search. Passengers wait at the door when the checkpoint is full, so source blocking shows how congested the checkpoint is.

Generated 2026-10-13 19:48 | simulated 4.0 h (warm-up 15 min) | seed 5 | 10 replications

HEALTH SCORE

89 /100

Grade B

Good: sound design with minor weaknesses worth addressing.

Critical 0 | High 1 | Medium 1 | Low 0 | Info 1

The score starts at 100 and loses points for each finding. The deductions are listed in Appendix B.

THROUGHPUT	COMPLETED	ARRIVALS LOST
476.8 units/h	1910 units	0.0%

AVERAGE WIP	CYCLE TIME	BOTTLENECK
22.3 units	168 s	xray_1

Executive summary

Primary bottleneck: 'xray_1' (78% busy, 22% down, index 105). Sources are held back up to 5% of the time. Run what-if to measure which lever recovers the most throughput.

What to fix first

- High (xray_1): Add parallel capacity, speed the station up, or reduce downtime. Measured best lever: 10% faster processing at 'xray_2' (+1.2% throughput).
- Medium (xray_1): Shorten repairs (spares, on-call technicians) or introduce preventive maintenance. A 20% shorter repair time is a what-if lever.

Best measured improvement

Best lever: 10% faster processing at 'xray_2' -> +5.7 units/hour (1.2%), measured over 3 seeds with common random numbers.

Findings

Each finding names where the problem is, why it matters, and what to do. Severity reflects the impact on throughput, correctness or confidence in the results.

SEVERITY	AREA	WHERE	ISSUE AND RECOMMENDED ACTION	POINTS
High	Capacity	**xray_1**	The bottleneck is 100% utilised (busy or down). There is no slack: any disturbance directly cuts output. Action: Add parallel capacity, speed the station up, or reduce downtime. Measured best lever: 10% faster processing at 'xray_2' (+1.2% throughput).	-8

SEVERITY	AREA	WHERE	ISSUE AND RECOMMENDED ACTION	POINTS
Medium	Reliability	**xray_1**	Down 22% of the time (5 failures). Action: Shorten repairs (spares, on-call technicians) or introduce preventive maintenance. A 20% shorter repair time is a what-if lever.	-3
Info	Capacity	**xray_1**	Constraint of the line: 100% utilised. Action: Improvements elsewhere will not raise throughput until this station is relieved. Measured best lever: 10% faster processing at 'xray_2' (+1.2% throughput).	

Performance by node

Share of the measured window each node spent busy, blocked (finished part cannot leave), down (failed) and starved (waiting for work).

Legend: green = busy, amber = blocked, red = down, grey = starved or idle.

NODE	KIND	STATE MIX	BUSY %	BLOCKED %	DOWN %	AVG / MAX WIP
pax	source	-	-	-	-	-
fast_track	machine		35.9	0.0	0.0	0.5 / 4
id_1	machine		88.9	6.3	0.0	2.9 / 7
id_2	machine		72.5	6.4	0.0	2.4 / 7
belt	delay		45.2	45.2	0.0	5.5 / 14
xray_1 bottleneck	machine		78.3	0.0	21.6	3.6 / 5
xray_2	machine		98.0	0.0	0.0	3.4 / 5
xray_3	machine		81.5	0.0	6.6	3.1 / 5
secondary	machine		51.5	0.0	0.0	0.8 / 4
hall	sink	-	-	-	-	-

Work in process over time

Reliability and robustness

The pipeline was simulated 10 times with different random seeds. A wide interval means the result depends strongly on luck.

METRIC	MEAN	95% CONFIDENCE	STD DEV	RELATIVE
Throughput	503.09 units/h	+- 8.53	11.92	1.7%
Average WIP	21.17 units	+- 3.60	5.03	17.0%
Cycle time	151.32 s	+- 25.72	35.96	17.0%
Arrivals lost	0.00 %	+- 0.00	0.00	0.0%

No replication deadlocked.

Failure model

NODE	MTBF	MTTR	AVAILABILITY	OBSERVED DOWN	FAILURES
xray_1	90 min	5.0 min	94.7%	21.6%	5
xray_2	90 min	5.0 min	94.7%	0.0%	0
xray_3	90 min	5.0 min	94.7%	6.6%	3

Single points of failure

Every station has a parallel alternative: no single station can stop the whole pipeline.

Model validity checks

CHECK	RESULT	DETAIL
Unit conservation	PASS	created 1933 = completed 1910 + scrapped 0 + lost 0 + in system 23
Little's Law	PASS	L = 22.31, lambda x W = 22.23 (gap 0.3%)
Run length	PASS	window = 80 x cycle time (30x or more recommended)
Warm-up	PASS	first 15 min excluded from statistics
Deadlock-free	PASS	reference run completed

What-if analysis

Each lever was applied to one of the most constrained stations and the whole pipeline re-simulated over at least 8 hours with 3 identical random seeds per scenario. Baseline throughput: 494.4 units/h.

NODE	CHANGE	UNITS/H	CHANGE	CHANGE %
xray_2	10% faster processing	500.1	**+5.7**	+1.2%
xray_3	10% faster processing	498.9	**+4.6**	+0.9%
xray_1	10% faster processing	497.8	**+3.5**	+0.7%
xray_1	20% shorter repairs	496.9	**+2.6**	+0.5%

NODE	CHANGE	UNITS/H	CHANGE	CHANGE %
xray_2	20% shorter repairs	496.2	**+1.8**	+0.4%
xray_3	20% shorter repairs	495.4	**+1.0**	+0.2%
xray_1	+2 buffer slots	494.8	**+0.5**	+0.1%
xray_2	+2 buffer slots	494.8	**+0.5**	+0.1%
xray_3	+2 buffer slots	494.8	**+0.5**	+0.1%

Best lever: 10% faster processing at 'xray_2' -> +5.7 units/hour (1.2%), measured over 3 seeds with common random numbers.

Appendix A: node inventory

Effective parameters after node-type templates are applied. Parameters not listed use the node type's defaults.

NODE	TYPE	EFFECTIVE PARAMETERS
pax	source	interarrival=exponential(7), mix={"priority":0.2,"standard":0.8}, on_blocked=wait
fast_track	machine	capacity=5, process=normal(14,4)
id_1	machine	capacity=6, process=normal(15,4)
id_2	machine	capacity=6, process=normal(15,4)
belt	delay	capacity=14, transit=uniform(14,8)
xray_1	machine	capacity=4, mtbf=5400, mttr=300, process=normal(24,6)
xray_2	machine	capacity=4, mtbf=5400, mttr=300, process=normal(24,6)
xray_3	machine	capacity=4, mtbf=5400, mttr=300, process=normal(24,6)
secondary	machine	capacity=4, process=normal(70,20)
hall	sink	(defaults)

Appendix B: how the score is computed

Health score = 100 minus the sum of the penalties below, never less than 0. Grades: A 90 and above, B 80 to 89, C 70 to 79, D 60 to 69, F below 60. Penalties are capped per category so that one repeated issue cannot dominate the score. Informational findings carry no penalty.

SEVERITY	WHERE	FINDING	POINTS
High	**xray_1**	The bottleneck is 100% utilised (busy or down). There is no slack: any disturbance directly cuts output.	-8
Medium	**xray_1**	Down 22% of the time (5 failures).	-3
	Total deductions		**-11**
	Health score		**89**

Method: the pipeline is simulated as a discrete-event system; the bottleneck index is utilisation (busy plus down time) with small weights for starved downstream and blocked upstream neighbours. What-if results are measured, not estimated. Confidence

intervals use Student's t over independent replications.